Route G-R9b: Paired Exchange Channels, an Exact 4/9 Bound, and Two Ways to One Outlet

A side branch of G-R9, alongside G-R10; no mass, time-law or flux fit

9 October 2026

Abstract

G-R9 found that a fixed vertex contact between two identical three-node rings opens a second, degenerate exchange outlet. Here we show that this is structural. Flipping the sign of every link maps L to $4 - L$; on a three-node ring this pairs the lower gap of one ring with the upper gap of its mirror partner. Hence *every* resonant pair of flux 3-rings has the reverse outlet degenerate, and every real density–density contact couples the two outlets with equal strength. For the rank-one vertex contact, two exact dark states make the single-outlet probability at most 4/9 at every update and every contact strength, which G-R9 had stated only as a weak-contact envelope. One outlet is reached in two ways. A 3-ring paired with a 4-ring, joined by the same vertex contact, leaves only the target degenerate and transfers with probability $1 - O(\gamma^2)$. Two identical rings joined by an *oriented* contact form an exactly closed chiral triangle that transfers with unit probability. In both, the two copies must be distinguishable. Ordered updates, quantum mechanics, the resonant flux and the contact orientation remain inputs.

1 What G-R9 left open

G-R9 used $L_\theta = 2I - e^{\mathrm{i}\theta/3}S - e^{-\mathrm{i}\theta/3}S^\dagger$ on two copies, read clock A on modes $(0,1)$ and clock B on modes $(1,2)$, and asked for the exchange $|1,1\rangle \to |0,2\rangle$. At $\theta = \pi/2$ the free gaps are both $\sqrt{3}$. The contact $\gamma P_0 \otimes P_0$ then also reaches $|2,0\rangle$, which has the same energy. Swap symmetry gave an exact bound $P_{11\to 02} \le \frac{1}{2}$; a weak-contact block gave the envelope 4/9, which was explicitly *not* asserted for the full generator. Its next bounded choice was either to keep all three channels or to find an independent reason that removes the extra outlet. G-R10 took the first branch: with the contact unchanged it built the all-channel readout, found two exact dark states and a conserved 2/3 dark weight of $|1,1\rangle$. This side note, written in parallel and not replacing G-R10, asks whether the extra outlet is an accident of identical copies at $\theta = \pi/2$, and gives one construction for one-outlet exchange in each direction of that choice.

Notation. On an N-node ring, $S|j\rangle = |j+1 \bmod N\rangle$ and

$$L_N(\theta) = 2I - e^{\mathrm{i}\theta/N}S - e^{-\mathrm{i}\theta/N}S^\dagger, \qquad \lambda_k^{(N)}(\theta) = 2 - 2\cos\frac{\theta - 2\pi k}{N}, \tag{1}$$

with flux-independent Fourier eigenvectors $f_k(j) = e^{2\pi\mathrm{i}jk/N}/\sqrt{N}$. Labels a, b below are energy-sorted modes of each copy. For a 3-ring with $0 < \theta < \pi$ the sorted order is $k = 0, 1, 2$. The input is $|1,1\rangle$, the *target* outlet $|0,2\rangle$ and the *reverse* outlet $|2,0\rangle$.

2 A link sign flip pairs the channels

Lemma 1 (link sign flip). $4I - L_N(\theta) = L_N(\theta + N\pi)$; $L_N(\theta + 2\pi) = DL_N(\theta)D^\dagger$ with $D = \mathrm{diag}(e^{2\pi\mathrm{i}j/N})$; and $L_N(\theta)^* = L_N(-\theta)$. *Proof.* Replacing every link amplitude by its negative adds π to each of the N link phases, so the loop phase grows by $N\pi$. The other two identities follow from $DSD^\dagger = e^{2\pi\mathrm{i}/N}S$ and $S^* = S$. □

For even N the shift $N\pi$ is a multiple of 2π, so the spectrum is symmetric about 2 at every flux (the ring is bipartite). For $N = 3$ the shift is π modulo 2π, and with conjugation

$$\operatorname{spec} L_3(\pi - \theta) = 4 - \operatorname{spec} L_3(\theta). \tag{2}$$

The sorted gaps follow in closed form:

$$\Delta_-(\theta) = \lambda_1 - \lambda_0 = 2\sqrt{3}\,\sin\frac{\pi-\theta}{3}, \qquad \Delta_+(\theta) = \lambda_2 - \lambda_1 = 2\sqrt{3}\,\sin\frac{\theta}{3}. \tag{3}$$

Theorem 1 (channel pairing). Let A and B be 3-rings with fluxes $\theta_A, \theta_B \in (0, \pi)$. Then $E_{02} = E_{11}$ holds if and only if $\theta_B = \pi - \theta_A$, and in that case also $E_{20} = E_{11}$. *Proof.* $E_{02} = E_{11}$ means $\Delta_-(\theta_A) = \Delta_+(\theta_B)$. By Eq. (3) this is $\sin\frac{\pi-\theta_A}{3} = \sin\frac{\theta_B}{3}$ with both arguments in $(0, \pi/3)$, so $\theta_B = \pi - \theta_A$. Then $E_{20} - E_{11} = \Delta_+(\theta_A) - \Delta_-(\pi - \theta_A) = 0$. □

So no choice of the two fluxes separates the outlets. G-R9's identical copies at $\pi/2$ are the self-mirror case of this family. Any flux is gauge- or conjugation-equivalent to one in $[0, \pi]$, and at the ends a ring level is itself degenerate. The pairing is special to three levels: for $N > 3$ the flip pairs gap $(0, 1)$ with gap $(N-2, N-1)$, not with the middle dial. The verifier finds the resonant partner by bisection on 49 values of θ_A and confirms $\theta_B = \pi - \theta_A$ and $E_{20} = E_{11}$ to 10^{-10}.

3 What a local contact can and cannot see

Take a real density–density contact $V = \sum_{j,l} c_{jl}\, P_j \otimes P_l$ between two 3-rings. In the mode basis

$$\langle a, b|V|a', b'\rangle = \hat{c}(a' - a,\, b' - b), \qquad \hat{c}(p, q) = \frac{1}{9}\sum_{j,l} c_{jl}\,\omega^{jp+lq}, \quad \omega = e^{2\pi\mathrm{i}/3}. \tag{4}$$

Lemma 2 (equal coupling). On the resonant triple $(|0,2\rangle, |1,1\rangle, |2,0\rangle)$ every diagonal entry is $c_0 = \hat{c}(0,0)$, and $V_{02,11} = V_{11,20} = V_{20,02} = z \equiv \hat{c}(1,-1)$. The block is therefore circulant, $c_0 I + zC + z^* C^\dagger$: a three-site ring whose loop phase is $\Phi = 3\arg z$. Both outlets couple to the input with the same magnitude $|z|$. If c is symmetric, z is real and $\Phi \in \{0, \pi\}$. *Proof.* The three elements $V_{02,11}$, $V_{11,20}$ and $V_{20,02}$ all carry momentum transfer $(1, -1)$ modulo 3. Reality of c gives $\hat{c}(-p, -q) = \hat{c}(p, q)^*$, so $V_{20,11} = z^*$. □

With real z the block is $c_0 - z + 3z|T\rangle\langle T|$, with $|T\rangle = (|0,2\rangle + |1,1\rangle + |2,0\rangle)/\sqrt{3}$. In the weak-contact block this is G-R9's 4/9 structure, now for every symmetric contact and every mirror pair. For identical rings a symmetric c also makes the full generator swap-symmetric, so G-R9's exact bound $P \le \frac{1}{2}$ covers all of them.

Lemma 3 (rank-one dark states). Let $H = H_0 + \gamma|v\rangle\langle v|$ and let $\mathcal{D}_0$ be the eigenspace of H_0 at E_0, containing $|i\rangle \neq |t\rangle$. Put $|b\rangle = P_{\mathcal{D}_0}|v\rangle/\|P_{\mathcal{D}_0} v\|$, $w_x = |\langle x|b\rangle|^2$ and $G(s) = \langle b|e^{-\mathrm{i}Hs}|b\rangle$. Then for every s

$$\langle t|e^{-\mathrm{i}Hs}|i\rangle = \langle t|b\rangle\langle b|i\rangle\big[G(s) - e^{-\mathrm{i}E_0 s}\big], \qquad P_{i\to t}(s) \le 4 w_i w_t. \tag{5}$$

Proof. Vectors of $\mathcal{D}_0$ orthogonal to $|b\rangle$ are also orthogonal to $|v\rangle$, so they are exact eigenvectors at E_0; their orthogonal complement is invariant and contains $|b\rangle$. Write $|i\rangle = \langle b|i\rangle|b\rangle + |d_i\rangle$ and use $\langle t|d_i\rangle = -\langle t|b\rangle\langle b|i\rangle$. The bound follows from $|G| \le 1$. □

The vertex contact is rank one with all $|v_{ab}|^2 = 1/9$. For identical rings or any mirror pair, $w = 1/3$ on the triple, so

$$P_{11\to 02}(s) = P_{11\to 20}(s) = \tfrac{1}{9}\big|G(s) - e^{-\mathrm{i}E_0 s}\big|^2 \le \tfrac{4}{9} \tag{6}$$

exactly, for every γ and every update. The two outlets have identical amplitudes even for mirror pairs, which lack swap symmetry. Here $|1,1\rangle$ has bright weight 1/3, the complement of G-R10's conserved dark weight 2/3. For weak contact $G \simeq e^{-\mathrm{i}(E_0 + \gamma/3)s}$, which reproduces G-R9's $\frac{4}{9}\sin^2(\gamma s/6)$

and G-R10's weak-contact amplitude; Eq. (5) makes that amplitude exact once the exact G is used. The verifier checks Eq. (5) against full 9×9 evolution, and computes G independently from the roots of the secular equation $1 = \gamma \sum_{ab} |v_{ab}|^2/(E - \varepsilon_{ab})$.

θ_A	γ	max sampled P_{02}	bound
$\pi/2$	0.01	0.444444088	4/9
$\pi/2$	0.5	0.439608352	4/9
$\pi/2$	10	0.417446514	4/9
$\pi/3$ (B at $2\pi/3$)	2	0.386793114	4/9

Equation (5) also says what unit transfer through a rank-one contact requires: $w_i = w_t = \frac{1}{2}$. The input's energy shell must contain exactly one other state that the contact touches.

4 Exit A: distinguishable spectra

Keep the G-R9 vertex contact and replace copy B by a 4-ring. For $0 < \theta < \pi$ its sorted spectrum is $2 \mp 2\cos(\theta/4)$, $2 \mp 2\sin(\theta/4)$, so its middle gap is $4\sin(\theta/4)$. With A at $\theta_A = \pi/2$ ($\Delta_- = \sqrt{3}$), resonance fixes

$$\theta_B = 4 \arcsin \frac{\sqrt{3}}{4} = 1.791329588, \qquad \operatorname{spec} L_4(\theta_B) = \Big\{2 \mp \frac{\sqrt{13}}{2},\ 2 \mp \frac{\sqrt{3}}{2}\Big\}. \tag{7}$$

Among the twelve product levels, the shell of $|1,1\rangle$ ($E_0 = 4 - \sqrt{3}/2$) contains only $|1,1\rangle$ and $|0,2\rangle$. The nearest other level is the reverse outlet, detuned by $(3\sqrt{3} - \sqrt{13})/2 = 0.7953$.

This holds across the whole resonant domain. A level $|a,b\rangle$ is degenerate with $|1,1\rangle$ iff $\lambda_a - \lambda_1 = \mu_1 - \mu_b$. Since B is non-degenerate, $a = 1$ forces $b = 1$ and $a = 0$ forces $b = 2$; only $a = 2$, $b = 0$ could add a state. On a 2001-point grid of $\theta_A \in (0.2756, \pi)$, where the lower end is the flux at which Δ_-^A equals the largest possible 4-ring middle gap $2\sqrt{2}$, the reverse detuning never falls below 0.318.

Now Lemma 3 has $w_i = w_t = \frac{1}{2}$ and no bound below one: $P_{11\to 02} = \frac{1}{4}|G - e^{-\mathrm{i}E_0 s}|^2$. The bright pair $(|1,1\rangle + |0,2\rangle)/\sqrt{2}$ is shifted by $\gamma/6$ while $(|1,1\rangle - |0,2\rangle)/\sqrt{2}$ is an exact eigenstate, so for weak contact $P \simeq \sin^2(\gamma s/12)$. With $\alpha = 1/4$ and integer updates, the full 12×12 evolution gives:

γ	best n	P_{02}	P_{20}	$(1 - P_{02})/\gamma^2$
0.09	856	0.999543095	9.7×10^{-5}	0.056
0.03	2535	0.999936858	4.3×10^{-7}	0.070
0.01	7558	0.999994927	3.1×10^{-7}	0.051

The best update is later than the weak estimate $6\pi/(\gamma\alpha)$ by about 2%, 0.9% and 0.2%: the exact bright root is $E_0 + \frac{\gamma}{6}(1 - 0.196\gamma + \dots)$, so the exchange slows slightly. The deficit is $O(\gamma^2)$ and comes from dressing by off-shell levels. The reverse outlet is not removed by selecting an operator: the partner's spectrum detunes it, and the contact is unchanged. What is still supplied is one tuned flux, as at G-R9's $\theta = \pi/2$.

5 Exit B: an oriented contact on identical rings

Keep two identical rings and orient the contact:

$$V_h = \gamma \sum_{j,l} h\big((j - l) \bmod 3\big)\, P_j \otimes P_l. \tag{8}$$

It commutes with the joint translation $S \otimes S$, which conserves $a + b \bmod 3$. The class $a + b \equiv 2$ is exactly the resonant triple, so the triple is a closed sector for every γ. Unlike the vertex contact,

there is no leakage. Here $c_0 = \gamma(h_0 + h_1 + h_2)/3$ and $z = \frac{\gamma}{3}(h_0 + h_1\omega + h_2\omega^2)$. Swapping the copies sends $h(d) \to h(-d)$, so a swap-symmetric profile ($h_1 = h_2$) has real z, $\Phi = 0$, and stays at 4/9.

The profile $h = (2, 0, 1)$ links node j of A to node j of B with weight 2γ and to node $j + 1$ with weight γ. It gives $z = (\gamma/\sqrt{3})e^{-i\pi/6}$ and $\Phi = -\pi/2$. The circulant then has equally spaced eigenvalues $c_0 + \{\sqrt{3}, 0, -\sqrt{3}\}|z|$, and

$$e^{-iHt_*}|1,1\rangle = e^{i\varphi}|0,2\rangle, \qquad t_* = \frac{2\pi}{3\gamma}, \tag{9}$$

for some phase φ. This is the chiral quantum walk on a triangle: a loop phase of $\pm\pi/2$ makes transport directional [2]. The reversed orientation $h = (2, 1, 0)$ sends the quantum to $|2, 0\rangle$ instead. The same holds for any mirror pair. Sampled at integer updates ($\alpha = 1/4$):

γ	t_*	n	P_{02} at n	P_{20} at n
0.09	23.271	93	0.9999976057	1.2×10^{-6}
0.03	69.813	279	0.9999976057	1.2×10^{-6}
0.01	209.440	838	0.9999997561	1.2×10^{-7}

The deficit is only the offset between t_* and the nearest update; at t_* itself the full 9×9 value is one to 10^{-10}. Here the third channel is not removed. The oriented contact routes the quantum around the resonant triangle, and all three channels stay in the model. What is supplied is an orientation and the weight ratio that sets $\Phi = \pm\pi/2$. That value gives exact transfer at t_*; it is not claimed to be the only value that can give high transfer at later times.

Exit B does *not* remove G-R10's preparation obstruction. Any translation-covariant contact makes the closed triple circulant, so $|T\rangle$ is an exact eigenvector and the $|T\rangle$ weight 1/3 of $|1,1\rangle$ is conserved. G-R10's dark clock state $|D_+\rangle = (|0,2\rangle - 2|1,1\rangle + |2,0\rangle)/\sqrt{6}$ is orthogonal to $|T\rangle$, so its fidelity from $|1,1\rangle$ never exceeds 2/3, for every profile h and every time. The verifier checks this for the oriented, reversed, symmetric and four random profiles.

6 What changed, and the next bounded step

copies and contact	states at E_{11}	best single outlet	status
identical, vertex (G-R9)	3	$\leq 4/9$	exact, all γ
mirror pair, vertex	3	$\leq 4/9$	exact, all γ
identical, any symmetric c	3	$\leq 1/2$	exact (swap)
3-ring + 4-ring, vertex (A)	2	$1 - O(\gamma^2)$	full 12×12
identical, oriented (B)	3	1 at t_*	exact closed sector

Physical reading. A one-for-one exchange between two dials needs the copies to be distinguishable, either in their spectra (Exit A) or in how their contact is oriented (Exit B). Identical 3-rings, or any mirror pair, joined by an unoriented local contact split the quantum between two outlets: exactly for identical copies and for the vertex contact, and in the weak-contact block otherwise. In the language of Chen's Sec. 1.9 [1], exchange between compatible event rhythms becomes one-to-one only when the partners differ in some respect. This is our inference for this toy model, not a statement of the source.

Still assumed. Quantum kinematics, the update order n, the common α, the preparation and the readout are inputs, as in G-R9. Exit A tunes one flux; Exit B chooses an orientation and a ratio. No dynamical principle selects either. No physical seconds, records, metric or particle spectrum follow.

Relation to G-R10 and next step. The two notes agree: the dark states behind the exact 4/9 bound are the same exact dark states that G-R10 uses as a stationary conditional clock. Exits A and B are declared alternatives for one-outlet *exchange* only. Neither prepares G-R10's $|D_+\rangle$, so G-R10's own next choice (a declared preparation and record instrument, or the unfiltered stationary mixture) is unchanged. Exit B is an exactly closed, solvable three-channel exchange on which G-R10's readout could be repeated. Separately, a spatial carrier would need a gapless network rather than another internal ring; that is not attempted here.

Executable verification. The companion program is `verify_gr9b_paired_channels.py`. Its inputs are fixed in `PAIRED_CHANNEL_CARD.json`, and it uses numpy only. Its 586 checks cover: Lemma 1 as matrix identities for $N = 3, 4$; the mirror (2) and gaps (3); Theorem 1 by independent bisection; Lemma 2 on 12 random real contacts; Lemma 3 against full evolution and against the secular-equation roots, with the 4/9 bound at strong contact; the Exit-A census, domain scan, second-order bright root and 12×12 fixtures; and the Exit-B closure, transfer, orientation reversal and the 2/3 bound on $|D_+\rangle$. Propagators use eigendecomposition and are cross-checked by an independent Taylor scaling-and-squaring exponential. The G-R9 code, card and TeX are preserved by SHA-256. These are checks of finite algebra, not experiments.

References

[1] BoYu Chen, *Energy spectrum for elementary particles*, final version, 23 April 2022, section 1.9. Conceptual source only.

[2] Z. Zimborás, M. Faccin, Z. Kádár, J. D. Whitfield, B. P. Lanyon and J. Biamonte, Quantum transport enhancement by time-reversal symmetry breaking, Sci. Rep. 3, 2361 (2013); arXiv:1208.4049.